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II SEMESTER B.Tech.

MID-SEMESTER EXAMINATION, February 2024

Course Code- ECITC2O3/EIITC2O3
Course Title- Data Structure and Algorithms

Time- **1.5 Hrs** Max. Marks(M)- **20**

Note: - All questions are compulsory. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	What is Data Structure? List a few examples of different data structures.	2M	CO1
1b	What are sparse matrices?	2M	CO5
2a	Write algorithms for Push() and Pop() operations in stacks.	2M	CO2
2b	Convert the following arithmetic expression written in infix notation to postfix notation: **A+(B*C)/(D-E)**	2M	CO2
3a	Evaluate the given postfix expression- 5 | 6 | 2 | + | * | 1 | 2 | 4 | / | -	2M	CO2
3b	What is a queue data structure? Why is it called FIFO?	2M	CO2
4a	What are the different types of queues?	2M	CO2
4b	Why do we need linked lists?	2M	CO2
5a	Write an algorithm to traverse a singly linked list.	2M	CO4
5b	Write an algorithm to insert an element in a queue using a singly linked list.	2M	CO3